

The Over-Dispersed Skellam (ODS) Reserving Model

A Signed Stochastic Framework for Incurred Chain-Ladder Uncertainty

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Abstract

We introduce the over-dispersed Skellam (ODS) model as a conditional stochastic layer around a selected incurred chain-ladder (CL) completion when future incurred development may be positive or negative. Each future cell is generated as the difference of two independent gamma–Poisson mixtures. The fitted distribution preserves the selected CL mean in every future cell and therefore preserves the expected aggregate future incurred development and accident-year ultimates. Under the fitted one-sided sign restriction, positive and negative development ages may coexist, while each age has only one active signed component. The one-sided over-dispersed Poisson (ODP) model is a special case. The Mack mean square error of prediction (MSEP) calibration matches the fitted ODS aggregate process variance to the full Mack MSEP, including its estimation component. This is a match of one aggregate second moment only. An absolute-value Pearson-type calibration is also examined. In the stated simulation, all methods capture the mean adequately because they share the fitted CL centre, but their uncertainty assessments differ: the process-only benchmark under-covers, the Mack-MSEP calibration is close to nominal, and the Pearson calibration produces wider intervals and over-coverage. These findings are specific to the stated data-generating process and do not provide general coverage guarantees.

Keywords: claims reserving, chain-ladder, run-off triangles, mean squared error of prediction, MSEP, over-dispersed Poisson model, prediction uncertainty, negative incremental claims.

1 Introduction

Stochastic reserving methods quantify uncertainty around a deterministic reserve estimate. They supplement the point estimate with a predictive distribution ([England and Verrall, 2002, 2006](#);

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[Wüthrich and Merz, 2008](#); [Scarth *et al.*, 2020](#)). This task is less direct for incurred triangles than for paid triangles because case-reserve reassessments and recoveries can make incremental incurred development negative. This paper considers a conditional predictive distribution for future incurred development around a selected incurred CL completion. The target is future incurred movement in the triangle. It is separate from future payments, payment-based best estimates, reported but not settled (RBNS), and incurred but not reported (IBNR) calculations. Mapping the model to those quantities requires paid, case-reserve, recovery, and accounting information outside the present model.

The framework conditions on the selected observed triangle and could apply to e.g. annual, quarterly, and monthly development periods. Section 2 defines the selected CL completion and the fitted ODS distribution. Section 3 derives their moment consequences. The proposed ODS layer provides a signed discrete predictive distribution for incurred development. The model uses one active sign at each development period. Positive and negative development periods can coexist. Section 2 gives the gamma–Poisson construction and its fitting conditions. Signed incurred development is already treated by loglinear and linear CL models and by shifted-lognormal models ([Verrall and Li, 1993](#); [de Alba, 2006](#)). Quasi-likelihood formulations also show that a CL mean structure need not rely on a non-negative response distribution ([Renshaw and Verrall, 1998](#)). These approaches have different predictive distributions, information requirements, and mean anchors. Mack (1994) stresses that the stochastic model underlying the CL estimate must be distinguished from alternative statistical constructions. In particular, the over-dispersed Poisson and generalized linear model (GLM) formulations can reproduce CL estimates under maximum likelihood, while their likelihood and independence assumptions differ from Mack’s moment assumptions ([Mack, 1994](#); [Kuang *et al.*, 2009](#)).

The actuarial literature also shows that the usual accident-year independence assumption can be inadequate when calendar- or accounting-year effects act along triangle diagonals. This issue is studied in the ODP and CL framework by [Brydon and Verrall \(2009\)](#) and [Wüthrich \(2010\)](#), and more recently in a Mack-aligned model with external information by [Riegel \(2026\)](#). These models address dependence and inflation effects, whereas the present ODS layer deliberately conditions on their absence. A related recent contribution derives an analytical MSEF for a stochastic lift of the generalized Cape Cod (GCC) method ([Richman and Wüthrich, 2026](#)). However it does not provide a signed discrete predictive distribution for incurred incrementals.

Taken together, these strands leave a narrower modeling gap for the present purpose: a conditional predictive layer that retains a selected incurred CL mean surface, permits negative integer incrementals, specifies an explicit cell and aggregate distributions, and calibrates its process variance to a Mack MSEF target without treating that target as identifying the predictive distribution.

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The paper presents ODS as one construction for this purpose, rather than as a replacement for calendar-year, individual-claim, Bornhuetter–Ferguson (BF), or GCC reserving models. The paper has three contributions. It defines a fitted ODS distribution around a selected incurred CL completion. It gives a Mack MSEF calibration and an absolute-value Pearson calibration. It studies their predictive intervals in a controlled simulation. The model is intended as a predictive layer for incurred reserving. Any mapping to payments, RBNS, IBNR, or accounting quantities requires a separate reserve decomposition.

2 The model

We define the deterministic incurred CL anchor first and then place the ODS predictive layer around it. The notation also records the observed upper triangle and the future lower triangle.

2.1 Triangle and incurred CL anchor

Let X_{ij} be incremental incurred development for accident period $i = 1, \dots, I$ and development period $j = 0, \dots, J$. At valuation time I , define

$$\begin{aligned}\mathcal{I}_I &= \{(i, j) : 1 \leq i \leq I, 0 \leq j \leq J, i + j \leq I\}, \\ \mathcal{L}_I &= \{(i, j) : 1 \leq i \leq I, 0 \leq j \leq J, i + j > I\}.\end{aligned}$$

The set $\mathcal{I}_I$ is the observed upper triangle. The set $\mathcal{L}_I$ is the future lower triangle. For each accident period define the last observed development age

$$j_i^* := \min\{J, I - i\}, \quad i = 1, \dots, I.$$

Note that in the square-design special case, $j_i^* = I - i$. The observed data are $\mathcal{D}_I = (X_{ij})_{(i,j) \in \mathcal{I}_I}$ and $\mathcal{F}_I = \sigma(\mathcal{D}_I)$. All fitted predictive quantities below are conditional on $\mathcal{F}_I$. The target is aggregate future incurred development,

$$R = \sum_{(i,j) \in \mathcal{L}_I} X_{ij}.$$

For $i = 1, \dots, I$ and $j = 0, \dots, J$, define cumulative incurred development by

$$C_{ij} = \sum_{k=0}^j X_{ik}, \quad C_{i,-1} = 0.$$

The selected volume-weighted incurred CL factor is

$$\hat{f}_j = \frac{\sum_{i=1}^{I-j-1} C_{i,j+1}}{\sum_{i=1}^{I-j-1} C_{ij}}, \quad j = 0, \dots, J-1,$$

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provided the denominator is positive and finite. Note that negative incrementals are allowed. The standard Mack calculation requires positive cumulative denominators and positive selected factors. For $i = 1, \dots, I$, complete the selected CL triangle recursively over $j = 0, \dots, J - 1$ whenever $(i, j + 1) \in \mathcal{L}_I$:

$$\hat{C}_{ij}^{\text{CL}} = C_{ij} \quad ((i, j) \in \mathcal{I}_I), \quad \hat{C}_{i,j+1}^{\text{CL}} = \hat{f}_j \hat{C}_{ij}^{\text{CL}} \quad ((i, j + 1) \in \mathcal{L}_I).$$

For $i = 1, \dots, I$ and $j = 0, \dots, J$, its incremental means are

$$\begin{aligned} \hat{\mu}_{ij}^{\text{CL}} &:= \hat{C}_{ij}^{\text{CL}} - \hat{C}_{i,j-1}^{\text{CL}}, \quad \hat{C}_{i,-1}^{\text{CL}} = 0, \\ \hat{R}^{\text{CL}} &= \sum_{(i,j) \in \mathcal{L}_I} \hat{\mu}_{ij}^{\text{CL}} = \sum_{i=1}^I \left(\hat{C}_{iJ}^{\text{CL}} - C_{i,j_i^*} \right). \end{aligned}$$

2.2 Mack benchmark and calibration target

The Mack CL model is used here as the deterministic incurred-CL benchmark and as an aggregate calibration target. We state its three model assumptions separately, following [Mack \(1993\)](#):

Assumption 1 (Mack CL assumptions). For $i = 1, \dots, I$ and $j = 0, \dots, J - 1$:

(MCL1) The accident-period cumulative paths $(C_{i0}, \dots, C_{iJ})$ are mutually independent across i .

(MCL2) There exist development factors $f_j > 0$ such that

$$\mathbb{E}[C_{i,j+1} \mid C_{i0}, \dots, C_{ij}] = f_j C_{ij}.$$

(MCL3) There exist variance parameters $\sigma_j^2 > 0$ such that

$$\text{Var}(C_{i,j+1} \mid C_{i0}, \dots, C_{ij}) = \sigma_j^2 C_{ij}.$$

Together, MCL1–MCL3 define the benchmark at the accident-period level: MCL1 separates accident periods, while MCL2–MCL3 specify their conditional first two moments ([Buchwalder et al., 2006](#)). The selected CL completion supplies the ODS mean anchor, and the Mack MSEP supplies the aggregate calibration target.

Mack regularity conditions. For the CL and Mack formulas below, cumulative variables are non-negative, $I \geq J + 1$, and every selected development factor has at least one age-to-age pair. Whenever $C_{i,j+1}/C_{ij}$ is used, require $C_{ij} > 0$ for $i = 1, \dots, I - j - 1$ and $j = 0, \dots, J - 1$. Smaller triangles use the corresponding edge-case definitions; all other denominators and variance estimates are required to be positive and finite. For $j = 0, \dots, J - 2$, use

$$\hat{\sigma}_j^2 = \frac{1}{I - j - 2} \sum_{i=1}^{I-j-1} C_{ij} \left(\frac{C_{i,j+1}}{C_{ij}} - \hat{f}_j \right)^2.$$

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In the square design used in the numerical study, $I = J + 1$. The terminal age $j = J - 1$ then has exactly one observed link ratio. In this one-ratio case, use the standard Mack extrapolation

$$\hat{\sigma}_{J-1}^2 = \min \left\{ \hat{\sigma}_{J-2}^2, \hat{\sigma}_{J-3}^2, \frac{(\hat{\sigma}_{J-2}^2)^2}{\hat{\sigma}_{J-3}^2} \right\},$$

when $I = J + 1$, $J \geq 3$, and the preceding estimates are positive (Mack, 1993, 1999). For a rectangular array with $I \geq J + 2$, the terminal age has at least two observed link ratios, and its variance coefficient is instead estimated directly by

$$\hat{\sigma}_{J-1}^2 = \frac{1}{I - J - 1} \sum_{i=1}^{I-J} C_{i,J-1} \left(\frac{C_{iJ}}{C_{i,J-1}} - \hat{f}_{J-1} \right)^2.$$

This distinguishes the one-ratio square case from the rectangular case. The aggregate Mack MSEF used for calibration is

$$\widehat{\text{MSEF}}_{\text{Mack}}(R) = \underbrace{\sum_{i=1}^I (\hat{C}_{iJ}^{\text{CL}})^2 \sum_{j=j_i^*}^{J-1} \frac{\hat{\sigma}_j^2}{(\hat{f}_j)^2 \hat{C}_{ij}^{\text{CL}}}}_{\text{process uncertainty}} + \underbrace{\sum_{j=0}^{J-1} \frac{\hat{\sigma}_j^2}{(\hat{f}_j)^2 \sum_{\ell=1}^{I-j-1} C_{\ell j}} \left(\sum_{i=I-j}^I \hat{C}_{iJ}^{\text{CL}} \right)^2}_{\text{estimation uncertainty}}. \quad (1)$$

The target is used here as an aggregate prediction-error benchmark, while the ODS distribution treats the fitted parameters as plug-in values.

2.3 ODS model

The ODS layer retains the accident-period/development-period decomposition and represents a cell as a difference of two gamma-mixed Poisson variables. Let

$$\theta := (\alpha_1, \dots, \alpha_I, \beta_0^+, \dots, \beta_J^+, \beta_0^-, \dots, \beta_J^-).$$

For $\lambda_{ij}^\pm > 0$, write

$$\lambda_{ij}^\pm := \alpha_i \beta_j^\pm, \quad r_{ij}^\pm := \frac{\lambda_{ij}^\pm}{\varphi - 1}.$$

The ODS distribution has integer support, so all monetary amounts are recorded as integer multiples of the chosen monetary unit. The value of φ is then unit-dependent. The following ODS population and sampling specification defines the fitted distribution used below:

Assumption 2 (ODS assumptions).

(ODS1) *The triangle dimensions are finite, cells are integer-valued, $\varphi > 1$, $\alpha_i > 0$ for $i = 1, \dots, I$, and $\beta_j^\pm \geq 0$ for $j = 0, \dots, J$. Both signed components are permitted at the population level.*

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(ODS2) For every positive component intensity,

$$G_{ij}^{\pm} \sim \text{Gamma}\left(r_{ij}^{\pm}, \text{rate} = \frac{1}{\varphi - 1}\right).$$

If $r_{ij}^{\pm} = \lambda_{ij}^{\pm} = 0$, set $G_{ij}^{\pm} = 0$ almost surely.

(ODS3) The mixing variables $G_{ij}^{\pm}$ are mutually independent across cells and signs.

(ODS4) Conditional on all mixing variables, the Poisson variables are mutually independent and, for every positive component intensity,

$$Y_{ij}^{\pm} \mid G_{ij}^{\pm} \sim \text{Pois}(G_{ij}^{\pm}).$$

If $r_{ij}^{\pm} = \lambda_{ij}^{\pm} = 0$, set $Y_{ij}^{\pm} = 0$ almost surely.

(ODS5) The population is non-degenerate in the sense that

$$\sum_{j=0}^J (\beta_j^+ + \beta_j^-) > 0.$$

The factorisation is scale-invariant: the cell distributions identify the products $\lambda_{ij}^{\pm} = \alpha_i \beta_j^{\pm}$, not α_i and $\beta_j^{\pm}$ separately. The fitted construction below fixes this scale by the net-share normalisation

$$\sum_{j=0}^J (\hat{\beta}_{jI}^+ - \hat{\beta}_{jI}^-) = 1.$$

Remark 1 (Mack and ODS dependence assumptions). *MCL1 allows dependence among incremental cells within an accident period, whereas ODS3–ODS4 imply mutual independence of the cell components and hence of the cells conditional on the fixed parameters. Calendar-year effects act along triangle diagonals and can induce dependence across accident periods; their treatment is a natural extension of this development-level specification (Riegel, 2026). The present ODS distribution focuses on cellwise variation while retaining the practical ability to represent negative incrementals in incurred data (Riegel, 2026).*

Model 1 (The ODS reserving model). For accident period $i \in \{1, \dots, I\}$ and development period $j \in \{0, \dots, J\}$, let each cell X_{ij} denote incremental development. Under assumption 2, define each cell by $X_{ij} := Y_{ij}^+ - Y_{ij}^-$. Conditional on the mixing variables, the cell distribution is

$$X_{ij} \mid G_{ij}^+, G_{ij}^- \sim \text{Skellam}(G_{ij}^+, G_{ij}^-),$$

where $\text{Skellam}(0, 0)$ denotes the point mass at zero.

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Remark 2 (Negative-binomial convention). For $r > 0$ and $0 < p < 1$, define the negative-binomial (NB) distribution by

$$\mathbb{P}(Y = y) = \frac{\Gamma(y + r)}{\Gamma(r)y!} p^r (1 - p)^y, \quad y = 0, 1, \dots$$

It has mean $r(1 - p)/p$ and variance $r(1 - p)/p^2$. Integrating out the mixing variables gives

$$Y_{ij}^\pm \sim \text{NB}\left(r_{ij}^\pm, p = \frac{1}{\varphi}\right)$$

for positive $\lambda_{ij}^\pm$. Zero components are point masses at zero.

2.4 Fitted ODS construction and dispersion calibration

For the fitted predictive layer, all quantities defined below are deterministic functions of $\mathcal{D}_I$ and hence are $\mathcal{F}_I$ -measurable. We use φ for the population dispersion parameter and write the two data-driven calibrations as $\hat{\varphi}_{\text{Mack}}$ and $\hat{\varphi}_{\text{Pearson}}$. Thus, $\hat{\varphi}_I$ is the selected value of one of these calibrations,

$$\hat{\varphi}_I \in \{\hat{\varphi}_{\text{Mack}}, \hat{\varphi}_{\text{Pearson}}\}, \quad \hat{\varphi}_I > 1.$$

Let $\hat{\mathbb{P}}_I^{\text{ODS}}$ denote the fitted plug-in distribution for the future vector $(X_{ij})_{(i,j) \in \mathcal{L}_I}$ obtained by replacing (θ, φ) in model 1 by the fitted parameters defined below and $\hat{\varphi}_I$. The observed cells are held fixed at $\mathcal{D}_I$. We write $\hat{\mathbb{E}}_I$ and $\widehat{\text{Var}}_I$ for moments under this distribution. It is a plug-in approximation to the conditional distribution under the original model and uses the gamma-Poisson construction in model 1. Define the CL ultimates by $\hat{U}_i := \hat{C}_{ij}^{\text{CL}}$ for $i = 1, \dots, I$. The CL-to-ODS mapping is well-defined when

$$0 < \hat{U}_i < \infty, \quad 0 < |\hat{f}_j| < \infty, \quad i = 1, \dots, I, \quad j = 0, \dots, J - 1.$$

Define the development shares by

$$\hat{\gamma}_0 := \frac{1}{\prod_{k=0}^{J-1} \hat{f}_k}, \quad \hat{\gamma}_j := \frac{\hat{f}_{j-1} - 1}{\prod_{k=j-1}^{J-1} \hat{f}_k}, \quad j = 1, \dots, J.$$

For the Mack MSEP calibration, the Mack requirements impose $\hat{f}_j > 0$ and positive cumulative denominators. In addition, at least one future cell must have positive component intensity. Otherwise the Mack MSEP denominator is zero. The absolute-value condition belongs to the ODS CL-to-ODS mapping and is imposed in addition to the Mack requirements.

Here and below, an empty product equals one and an empty sum equals zero. For every future cell $(i, j) \in \mathcal{L}_I$, so that $j \geq j_i^* + 1$, the completed CL recursion gives

$$\hat{C}_{ij}^{\text{CL}} = \frac{\hat{U}_i}{\prod_{k=j}^{J-1} \hat{f}_k}.$$

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Consequently, $\hat{U}_i \hat{\gamma}_j$ is the completed CL increment at age j (whenever that age is used in the future triangle). The fitted ODS parameters are defined by

$$\hat{\alpha}_{iI} := \hat{U}_i, \quad \hat{\beta}_{jI}^+ := \max(\hat{\gamma}_j, 0), \quad \hat{\beta}_{jI}^- := \max(-\hat{\gamma}_j, 0).$$

Here $i = 1, \dots, I$ for $\hat{U}_i$ and $\hat{\alpha}_{iI}$, and $j = 0, \dots, J$ for $\hat{\gamma}_j$ and $\hat{\beta}_{jI}^\pm$.

Assumption 3 (Fitted one-sided sign structure). *The fitted predictive layer imposes $\hat{\beta}_{jI}^+ \hat{\beta}_{jI}^- = 0$ for $j = 0, \dots, J$. Thus, one signed component is active at each estimated development age. This fitted restriction concerns the estimated development shares. The population parameters in ODS1 of assumption 2 remain unrestricted.*

Consequently,

$$\hat{\mu}_{ij}^{\text{ODS}} := \hat{\alpha}_{iI} (\hat{\beta}_{jI}^+ - \hat{\beta}_{jI}^-) = \hat{U}_i \hat{\gamma}_j.$$

Because $\sum_{j=0}^J \hat{\gamma}_j = 1$, this normalisation makes $\hat{\alpha}_{iI}$ equal to the completed incurred CL ultimate. The completed CL recursion implies that $\hat{\mu}_{ij}^{\text{ODS}} = \hat{\mu}_{ij}^{\text{CL}}$ for future cells $(i, j) \in \mathcal{L}_I$. Thus the fitted ODS layer reproduces every future CL cell mean. To make the accident-year implication explicit, define the random completed ultimate for accident period i by

$$U_i^{\text{ODS}} := C_{i, j_i^*} + \sum_{j=j_i^*+1}^J X_{ij}, \quad \hat{U}_i^{\text{CL}} := \hat{C}_{ij}^{\text{CL}}.$$

Then, conditional on the observed triangle,

$$\hat{\mathbb{E}}_I[U_i^{\text{ODS}}] = C_{i, j_i^*} + \sum_{j=j_i^*+1}^J \hat{\mu}_{ij}^{\text{CL}} = \hat{U}_i^{\text{CL}}, \quad \hat{\mathbb{E}}_I[R] = \hat{R}^{\text{CL}}.$$

The fitted ODS process variance is derived in Section 3.

For the separate Pearson-type calibration, let $\hat{\mu}_{ij}^{\text{P}}$ denote the fitted observed-cell incremental mean from the age-to-age fit. In the simulation, this is the Mack CL fitted next-age cumulative value minus the observed current cumulative value. The future ODS distribution is determined independently of these observed-cell calibration means. The Mack MSEP in Equation (1) supplies an aggregate target. Its parameter-estimation component stays within the Mack target, while the ODS calibration adds process variation to the fitted plug-in distribution. We calibrate the fitted ODS process-variance scale using the future component-intensity scale:

$$\hat{\phi}_{\text{Mack}} = \frac{\widehat{\text{MSEP}}_{\text{Mack}}(R)}{\sum_{(i,j) \in \mathcal{L}_I} \hat{\alpha}_{iI} (\hat{\beta}_{jI}^+ + \hat{\beta}_{jI}^-)}, \quad (2)$$

provided the denominator is positive and the resulting calibration satisfies $\hat{\phi}_{\text{Mack}} > 1$. Under the fitted one-sided sign restriction in assumption 3, the denominator equals $\sum_{(i,j) \in \mathcal{L}_I} |\hat{\mu}_{ij}^{\text{CL}}|$. The

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coefficient is solely an aggregate calibration for the fitted plug-in distribution. Under the fitted ODS distribution and the one-sided development-period restriction, this choice gives

$$\widehat{\text{Var}}_{I,\text{proc}}(R) = \hat{\phi}_{\text{Mack}} \sum_{(i,j) \in \mathcal{L}_I} |\hat{\mu}_{ij}^{\text{CL}}| = \widehat{\text{MSEP}}_{\text{Mack}}(R),$$

whenever the calibration is admissible. This equality concerns one aggregate second moment. The Mack and ODS predictive distributions can still differ, as can their quantiles. Define the informative observed-cell set

$$\mathcal{I}_I^+ := \left\{ (i, j) \in \mathcal{I}_I : 0 < |\hat{\mu}_{ij}^{\text{P}}| < \infty, |X_{ij}| < \infty \right\}.$$

The Pearson-type dispersion estimator used in the numerical study is given below. Here, $|\mathcal{I}_I|$ denotes the number of observed incremental cells.

$$\hat{\phi}_{\text{Pearson}} = \frac{1}{\nu_{\text{P}}} \sum_{(i,j) \in \mathcal{I}_I^+} \frac{\left(|X_{ij}| - |\hat{\mu}_{ij}^{\text{P}}| \right)^2}{|\hat{\mu}_{ij}^{\text{P}}|}, \quad \nu_{\text{P}} = |\mathcal{I}_I| - (I + J). \quad (3)$$

Under the square-triangle assumption $J = I - 1$, this reduces to $\nu_{\text{P}} = |\mathcal{I}_I| - (2I - 1)$. The expression is a nominal degrees-of-freedom correction: it treats the first development column as conditioned upon and subtracts the J selected development factors. This estimator is reported only when $\nu_{\text{P}} > 0$ and $\mathcal{I}_I^+$ is nonempty and the resulting calibration satisfies $\hat{\phi}_{\text{Pearson}} > 1$. The numerator uses absolute-value residuals over informative observed cells. Unavailable fitted means are omitted, including the first development column, which is conditioned upon. A zero fitted mean is admissible only when the corresponding observed increment is also zero. A nonzero observation at a zero fitted mean makes the Pearson estimate undefined.

3 Structural properties

The following lemma derives the cell moments from the gamma-Poisson construction for $i = 1, \dots, I$ and $j = 0, \dots, J$, using the conditional cell mean μ_{ij} defined in Section 2.

Lemma 1 (Cell moments). *Under model 1 and in particular under ODS3–ODS4 in assumption 2, the cells are mutually independent and*

$$\mathbb{E}[X_{ij} \mid \theta, \varphi] = \alpha_i(\beta_j^+ - \beta_j^-) =: \mu_{ij}, \quad \text{Var}(X_{ij} \mid \theta, \varphi) = \varphi \alpha_i(\beta_j^+ + \beta_j^-).$$

Under a one-sided component parameterisation, $\text{Var}(X_{ij} \mid \theta, \varphi) = \varphi |\mu_{ij}|$.

Proof. For a positive component intensity λ , the gamma-Poisson mixture in model 1 is $\text{NB}(\lambda/(\varphi - 1), 1/\varphi)$ and therefore has mean λ and variance $\varphi\lambda$. The same identities hold for a zero component by the point-mass convention. Independence of the two components gives the displayed mean and variance for their difference. The last identity follows from $\beta_j^+ \beta_j^- = 0$. $\square$

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Recall from Section 2 that R is the aggregate future incurred development. The corresponding positive and negative component totals over the future triangle are denoted by $Y_{\mathcal{L}}^+$ and $Y_{\mathcal{L}}^-$, respectively.

Proposition 1 (Aggregate future distribution). *Define $\Lambda_{\mathcal{L}}^{\pm} := \sum_{(i,j) \in \mathcal{L}_I} \lambda_{ij}^{\pm}$, and $Y_{\mathcal{L}}^{\pm} := \sum_{(i,j) \in \mathcal{L}_I} Y_{ij}^{\pm}$. Then the two aggregate components are independent and $Y_{\mathcal{L}}^{\pm} \sim \text{NB}\left(\frac{\Lambda_{\mathcal{L}}^{\pm}}{\varphi-1}, p = \frac{1}{\varphi}\right)$ when $\Lambda_{\mathcal{L}}^{\pm} > 0$. A zero aggregate component is a point mass at zero. Consequently, $R = Y_{\mathcal{L}}^+ - Y_{\mathcal{L}}^-$ is a difference of two independent negative-binomial variables, with*

$$\mathbb{E}[R \mid \theta, \varphi] = \Lambda_{\mathcal{L}}^+ - \Lambda_{\mathcal{L}}^-, \quad \text{Var}(R \mid \theta, \varphi) = \varphi(\Lambda_{\mathcal{L}}^+ + \Lambda_{\mathcal{L}}^-).$$

Under a one-sided component parameterisation, $\Lambda_{\mathcal{L}}^+ + \Lambda_{\mathcal{L}}^- = \sum_{(i,j) \in \mathcal{L}_I} |\mu_{ij}|$. Conditional on all future mixing variables, the equivalent distribution is

$$R \mid (G_{ij}^{\pm})_{(i,j) \in \mathcal{L}_I} \sim \text{Skellam} \left(\sum_{(i,j) \in \mathcal{L}_I} G_{ij}^+, \sum_{(i,j) \in \mathcal{L}_I} G_{ij}^- \right).$$

The fitted distribution has the same aggregate representation after replacing unknown parameters by fitted values. In particular, by the construction in Section 2,

$$\hat{\mathbb{E}}_I[X_{ij}] = \hat{\mu}_{ij}^{\text{CL}}, \quad (i, j) \in \mathcal{L}_I,$$

and hence

$$\hat{\mathbb{E}}_I[R] = \hat{R}^{\text{CL}}, \quad \widehat{\text{Var}}_{I, \text{proc}}(R) = \hat{\varphi}_I \sum_{(i,j) \in \mathcal{L}_I} |\hat{\mu}_{ij}^{\text{CL}}|.$$

For the accident-year ultimate defined in Section 2,

$$\hat{\mathbb{E}}_I[U_i^{\text{ODS}}] = \hat{U}_i^{\text{CL}}, \quad i = 1, \dots, I.$$

The fitted ODS distribution preserves the selected CL mean surface and gives the aggregate process variance stated above. The aggregate representation supports two simulation methods. One method simulates future cells independently. The other simulates the two aggregate negative-binomial components directly.

Proposition 2 (ODP moment special case). *If $\beta_j^- = 0$ for $j = 0, \dots, J$, then for $i = 1, \dots, I$ and $j = 0, \dots, J$,*

$$\mathbb{E}[X_{ij} \mid \theta, \varphi] = \alpha_i \beta_j^+, \quad \text{Var}(X_{ij} \mid \theta, \varphi) = \varphi \alpha_i \beta_j^+,$$

which is the classical one-sided over-dispersed Poisson (ODP) mean-variance structure. ODP is therefore the one-sided special case of ODS ([England and Verrall, 2002](#)).

4 Numerical study

Appendix A specifies the complete data-generating process (DGP), which follows the ODS assumptions. The study evaluates prediction intervals for the aggregate future incurred development $R = \sum_{(i,j) \in \mathcal{L}_I} X_{ij}$, using the observed upper triangle to fit the incurred CL mean surface. Conditional on the fixed DGP parameters, the incremental cells are independent. Thus, the DGP is not a full Mack-generating model: Mack supplies conditional moment assumptions for the CL benchmark and MSEP calibration, but does not impose independence among increments within an accident period.

All three predictive distributions use the fitted incurred CL means and differ only in the dispersion used for future cells: the oracle process-only benchmark uses the known DGP dispersion, the Mack-MSEP method uses the aggregate Mack MSEP calibration, and the Pearson method uses the observed-cell Pearson calibration. Figure 1 provides a visual preview of the simulation design: the expected cumulative triangle and the corresponding CL development factors. The early negative development pattern reflects incurred reserve releases, followed by positive later development. The parameter values and cell-generation mechanism are given in Appendix A.

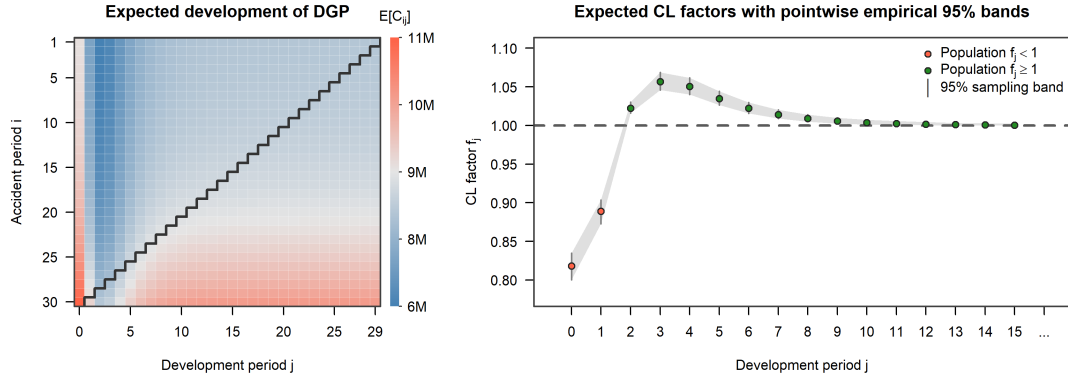


Figure 1: Visual overview of the simulation DGP: expected cumulative development and corresponding CL factors.

4.1 Simulation protocol

For each of $S = 10\,000$ outer replications, we simulate a full incremental triangle under Appendix A. We retain the observed cells in $\mathcal{I}_I$, regard the cells in $\mathcal{L}_I$ as future, and record the realised future reserve $R^{(s)}$. We then fit incurred Mack CL to the observed triangle and obtain its fitted future mean surface. The oracle process-only benchmark uses this fitted centre together with the known DGP dispersion φ_{DGP} ; it is infeasible because that dispersion is not estimated from the observed triangle. The two feasible alternatives use the same fitted centre with, respectively, the Mack-MSEP calibration $\hat{\varphi}_{\text{Mack}}$ and the observed-cell Pearson calibration $\hat{\varphi}_{\text{Pearson}}$.

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For each valid fitted replication and each of the three dispersion choices, we simulate $B = 10\,000$ conditionally independent future-cell realisations under the ODS distribution and sum the future cells to obtain reserve draws. A replication is valid only when both feasible dispersion estimates are finite and admissible. Equal-tailed empirical intervals use probabilities (2.5%, 97.5%) and (0.5%, 99.5%) for nominal 95% and 99% coverage, respectively; coverage is recorded when the interval contains the realised reserve $R^{(s)}$.

Algorithm 1 Monte Carlo protocol for coverage assessment

Input: Simulation design in Appendix A. Set $S = 10\,000$ and $B = 10\,000$.

- 1: **for** $s = 1, \dots, S$ **do**
 - 2: Simulate a full triangle, retain $\mathcal{I}_I$, and record $R^{(s)}$ over $\mathcal{L}_I$
 - 3: Fit incurred Mack CL and obtain the fitted future mean surface
 - 4: Calculate $\hat{\phi}_{\text{Mack}}^{(s)}$ and $\hat{\phi}_{\text{Pearson}}^{(s)}$
 - 5: **if** both feasible estimates are valid **then**
 - 6: **for** each $\phi \in \{\varphi_{\text{DGP}}, \hat{\phi}_{\text{Mack}}^{(s)}, \hat{\phi}_{\text{Pearson}}^{(s)}\}$ **do**
 - 7: Simulate B future reserves from the fitted ODS mean surface and dispersion ϕ
 - 8: Construct 95% and 99% intervals and record coverage indicators
 - 9: **end for**
 - 10: **end if**
 - 11: **end for**
 - 12: Report coverage and interval-width summaries over valid replications
-

4.2 Results

The point-estimation summaries appear in Table 1. The DGP oracle row reports the fixed DGP expected reserve and known dispersion; it is not a fitted method. The two feasible methods use the same incurred CL centre, so their average fitted reserves and average differences from the fixed DGP expected reserve are equal. The Mack-MSEP calibration uses the component-intensity scale over $\mathcal{L}_I$ to match the aggregate Mack MSEP. The Pearson calibration uses observed-cell residuals over $\mathcal{I}_I$ and targets a separate dispersion quantity. Neither method adds a separate parameter-draw layer: the fitted mean surface and calibrated dispersion are plug-in values in each outer replication.

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Method	Average reserve	Average centre error	Dispersion used
DGP oracle	7 322 061		100 000
Mack-MSEP	7 320 764	−1 297 (−0.02%)	172 417
Pearson-based	7 320 764	−1 297 (−0.02%)	7 851 377

Table 1: Method-wise reserve value, centre error relative to the fixed DGP expected reserve, and dispersion. For the feasible methods, values are averaged over valid simulated replications; the DGP oracle uses the fixed DGP expected reserve and known dispersion.

All predictive methods use the same fitted incurred CL mean surface. The common average fitted reserve is 7 320 764, close to the fixed DGP expected reserve of 7 322 061, with a difference of −0.02%. Thus, the mean centre is adequately captured in this simulation. However, the dispersion under Pearson calibration is materially larger than under the DGP oracle and Mack-MSEP calibrations.

Prediction-interval coverage. Figure 2 illustrates the empirical coverage and interval widths. All intervals are conditional on the fitted CL centre within an outer replication. Process-only intervals use the known DGP dispersion, so their inner draws represent future process variation but do not propagate uncertainty from estimating the dispersion or the fitted mean. Across outer replications, coverage nevertheless reflects both the displacement of the fitted centre and future process variation. In this experiment, process-only intervals under-cover, Mack-MSEP intervals are close to nominal, and Pearson intervals are much wider and over-cover. These are design-specific empirical coverage results, not unconditional claims about all incurred-reserving portfolios.

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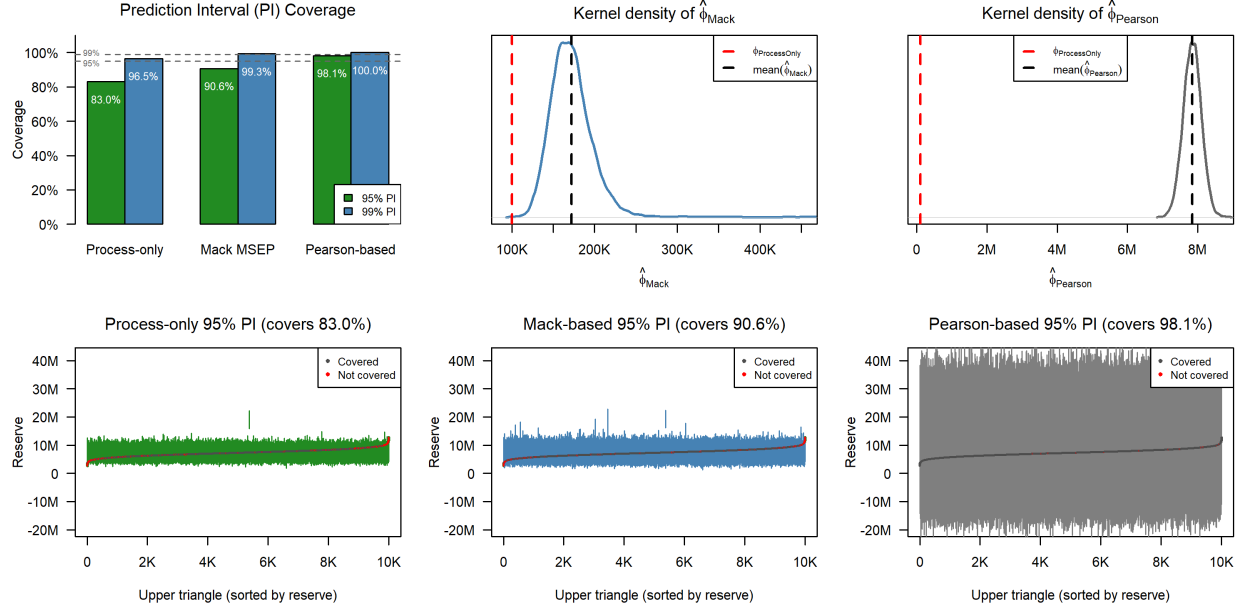


Figure 2: Distributions of the two feasible dispersion calibrations and 95% prediction intervals under process-only, Mack-MSEP, and Pearson-based dispersion.

5 Discussion

The numerical results describe calibration behaviour under the stated DGP from Appendix A. The process-only benchmark omits variation in the fitted CL completion, while the Mack MSEP calibration matches one aggregate second moment. Because the Mack assumptions specify conditional moments, the ODS and Mack distributions may differ in quantiles, tail probabilities, and other distributional features. The Pearson calibration targets a separate quantity and gives a distinct dispersion assessment. The combination of CL and Mack leaves multiple predictive distributions available. ODS is one construction that preserves the selected mean surface, supports signed development across ages, contains ODP as a special case, and permits consistent cellwise and aggregate simulation.

The same construction extends naturally to BF models. The selected CL increment means in Section 2 can be replaced by selected BF increment means. The Mack-MSEP calibration can be replaced by a BF MSEP estimator (Saluz, 2011). A related but distinct development is the GCC method: a recent preprint derives an analytical MSEP for a stochastic lift of GCC, addressing the gap between its deterministic formulation and prediction-error analysis (Richman and Wüthrich, 2026). GCC therefore provides a natural alternative mean-MSEP framework for future comparison, while ODS provides the signed discrete distribution considered here. The ODS moment results apply to any BF mean surface that satisfies the stated sign and admissibility conditions. This paper leaves numerical evaluation of that extension to future work.

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The conditional formulation could apply to both annual, quarterly, and monthly development periods. It can serve as a stochastic layer in direct best-estimate projection, latent IBNR modelling, and broader incurred-reserving applications. The relevant reserve, payment, case-reserve, and accounting decomposition must be specified separately.

Scope and extensions. The fitted one-sided sign restriction gives one active component at future development age, while both components remain available at the population level. The cell independence imposed by ODS3–ODS4 in assumption 2 focuses the model on development-level variation; calendar-year effects and common shocks would require an extension. The scalar dispersion multiplier imposes

$$\widehat{\text{Var}}_{I,\text{proc}}(X_{ij}) = \hat{\phi}_I |\hat{\mu}_{ij}^{\text{CL}}|, \quad (i, j) \in \mathcal{L}_I,$$

across fitted development periods. A development-specific extension can split an additional component intensity $\kappa_j \geq 0$ equally between the two fitted component shares:

$$\tilde{\beta}_j^+ = \hat{\beta}_{jI}^+ + \frac{\kappa_j}{2}, \quad \tilde{\beta}_j^- = \hat{\beta}_{jI}^- + \frac{\kappa_j}{2}, \quad j = 0, \dots, J.$$

With $\tilde{\lambda}_{ij}^\pm = \hat{\alpha}_{iI} \tilde{\beta}_j^\pm$, for $(i, j) \in \mathcal{L}_I$, the fitted cell means remain unchanged,

$$\tilde{\mu}_{ij} = \hat{\alpha}_{iI} (\tilde{\beta}_j^+ - \tilde{\beta}_j^-) = \hat{\mu}_{ij}^{\text{ODS}}, \quad (i, j) \in \mathcal{L}_I,$$

while the predictive variance becomes

$$\text{Var}_\kappa(X_{ij} \mid \mathcal{F}_I) = \hat{\phi}_I \hat{\alpha}_{iI} \left(\hat{\beta}_{jI}^+ + \hat{\beta}_{jI}^- + \kappa_j \right), \quad (i, j) \in \mathcal{L}_I.$$

Thus κ_j preserves the fitted means and increases the cell-level predictive variance by $\hat{\phi}_I \hat{\alpha}_{iI} \kappa_j$. It provides a development-specific alternative to a single constant over-dispersion multiplier while preserving the fitted mean surface. For $\kappa_j > 0$, both signed components are active and the cell can have two-sided integer support. Estimation and calibration of κ_j , together with calendar-year dependence and explicit parameter uncertainty, are natural extensions.

6 Conclusion

We develop ODS as a conditional predictive layer for future incurred development around a selected incurred chain-ladder completion. Whenever the fitting conditions are met, it preserves the selected CL mean in every future cell and therefore preserves the corresponding expected aggregate reserve and accident-year ultimates. The deterministic reserving centre is consequently separated from the stochastic distribution used to represent future variation.

The Mack MSEP calibration matches the aggregate second moment of the fitted ODS process variance. Distributional features, quantiles, and parameter-uncertainty treatment remain model-specific. The Pearson-type calibration targets a different quantity and is interpreted separately.

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In the stated simulation, all methods captured the mean adequately because they shared the same fitted CL centre, while their uncertainty assessments differed: the process-only benchmark under-covered, the Mack MSEP calibration was close to nominal, and the Pearson calibration produced wider intervals and over-covered. These findings are specific to the stated data-generating process and motivate validation under alternative portfolios and calibration choices.

ODS is therefore a transparent actuarial modelling option for conditional future incurred development, subject to portfolio-specific validation of its mean structure, sign restriction, dispersion calibration, and dependence assumptions. Its results provide model-based predictive assessments to be used alongside validation and actuarial judgement.

A Simulation study details

We introduce the DGP used for the numerical experiment. We set $I = J + 1 = 30$ and generate a full 30×30 incremental triangle with independent cells, as in ODS3–ODS4 of assumption 2. The cell mean is

$$\mu_{ij} = \alpha_i(\beta_j^+ - \beta_j^-), \quad i = 1, \dots, 30, \quad j = 0, \dots, 29. \quad (4)$$

The ultimate amounts α_i increase from 8 million to 10 million in the chosen monetary unit. Let $\gamma_j = \beta_j^+ - \beta_j^-$. We set

$$\gamma_0 = 1.1, \quad \gamma_1 = -0.2, \quad \gamma_2 = -0.1,$$

and allocate the remaining mass 0.2 over later development periods according to a discretised log-normal shape, so that $\sum_j \gamma_j = 1$. Define

$$\beta_j^+ = \max(\gamma_j, 0), \quad \beta_j^- = \max(-\gamma_j, 0).$$

Thus, development periods 1 and 2 have negative net mean increments.

For $\varphi_{\text{DGP}} = 100\,000$, each cell is generated as

$$X_{ij} = Y_{ij}^+ - Y_{ij}^-, \quad Y_{ij}^\pm \sim \text{NB} \left(\frac{\alpha_i \beta_j^\pm}{\varphi_{\text{DGP}} - 1}, p = \frac{1}{\varphi_{\text{DGP}}} \right), \quad (5)$$

with independent components and the point-mass convention for a zero component. The value of φ_{DGP} is tied to the chosen integer monetary unit. The DGP is therefore an instance of the ODS model, with negative early incurred development and positive later development. A visual overview is provided in Figure 1 in the numerical study.

Generative AI declaration

Generative AI was used for coding assistance and formatting. The author takes full responsibility for the final content.

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